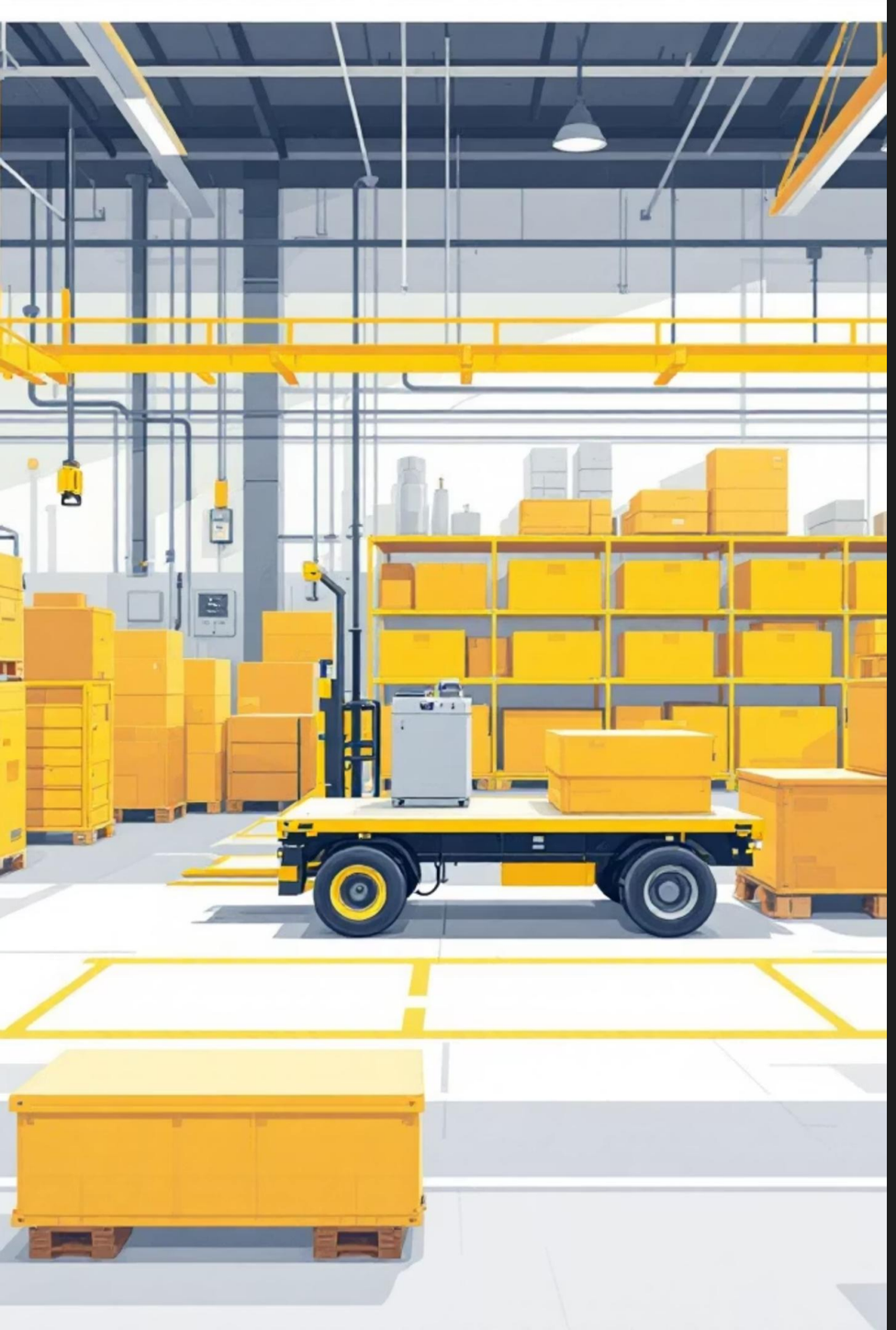

Material Handling Platform Trolley

Automated Engineering and Manufacturing Pipeline

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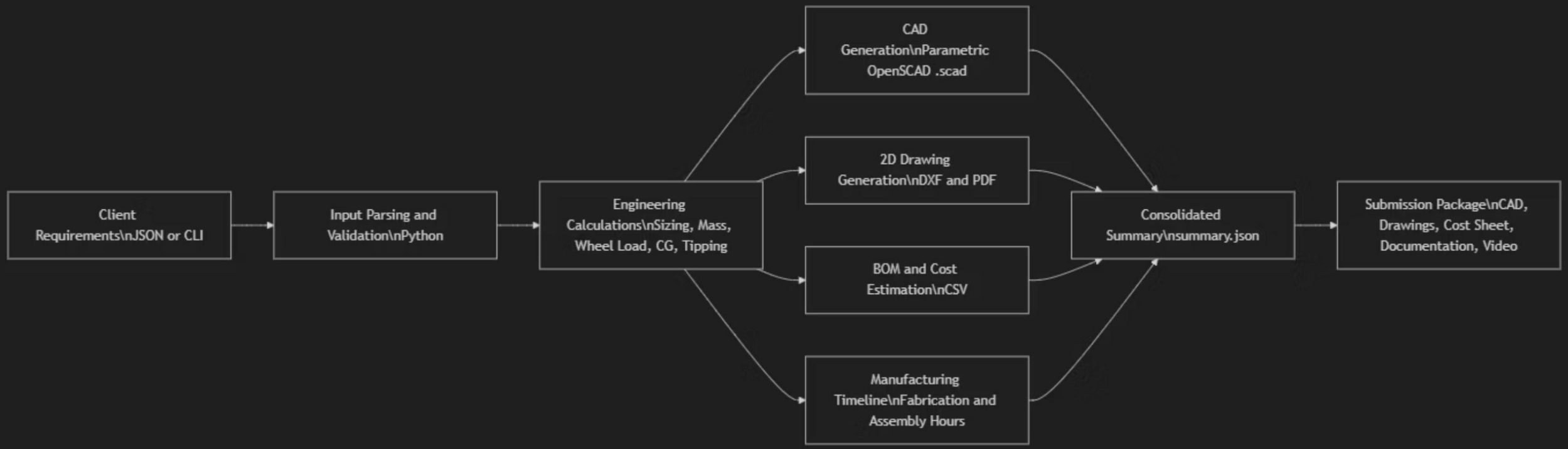
Detailed Process Workflow



This infographic illustrates our comprehensive workflow, highlighting key stages and decision points to ensure efficient project management

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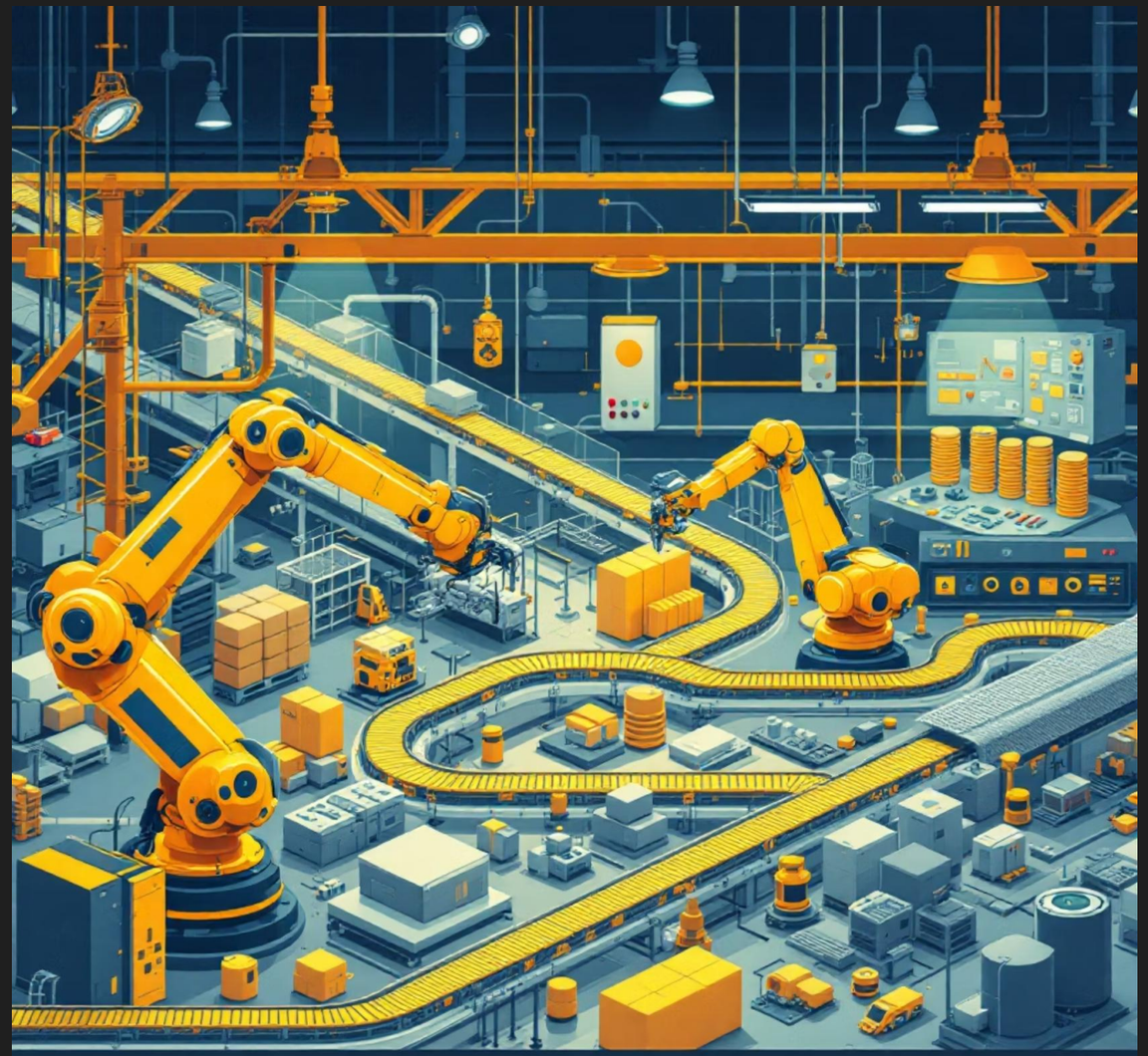
System Overview

This solution represents a modular Python pipeline that transforms client requirements into comprehensive engineering and manufacturing deliverables. The system bridges the gap between conceptual specifications and production-ready documentation through automated calculation, parametric CAD generation, and cost estimation.

The platform operates as an end-to-end workflow orchestrator, processing inputs through validation, engineering calculations, CAD generation, drawing production, bill of materials compilation, and timeline estimation. Each stage builds upon the previous, creating a cohesive package of technical documentation ready for manufacturing review.



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Workflow Architecture

The pipeline executes seven sequential stages, each producing validated outputs for downstream processing:

01

Requirement Parsing

Input validation and data model construction

02

Engineering Calculations

Structural and physical property determination

03

CAD Generation

Parametric 3D model script creation

04

Drawing Production

2D technical drawing generation

05

BOM and Costing

Material list and cost estimation

06

Timeline Estimation

Fabrication and assembly scheduling

07

Summary Generation

Consolidated deliverable package

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Data Flow Architecture

1. **Input Layer**
   JSON or CLI parameters
2. **Validation Engine**
   Data quality checks
3. **Calculation Core**
   Structural analysis
4. **CAD Generator**
   OpenSCAD scripts
5. **Drawing Engine**
   DXF and PDF output
6. **BOM Engine**
   Material costing
7. **Timeline Engine**
   Production scheduling
8. **Output Package**
   Complete deliverables

This linear data flow ensures traceability from requirements through to manufacturing documentation. Each processing stage validates inputs before proceeding, maintaining data integrity throughout the pipeline.

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Core Module Structure

models.py

Data models for input and output structures

calculations.py

Structural and physical property calculations

cad_generator.py

OpenSCAD parametric model generation

drawing_generator.py

DXF and PDF technical drawing production

bom_costing.py

Bill of materials and cost estimation

timeline.py

Fabrication and assembly timeline calculation

main.py

End-to-end workflow orchestration

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Technology Stack / Tools used

Python Ecosystem

The primary automation engine handles requirement parsing, calculation execution, file generation, costing logic, and workflow orchestration. Python's flexibility enables rapid development of calculation modules and seamless integration between disparate tools.

OpenSCAD Backend

Parametric CAD generation through dynamically written .scad scripts. Python constructs parameterized CAD code that OpenSCAD renders and exports as 3D models, enabling programmatic design from engineering specifications.

Matplotlib Visualization

Lightweight PDF drawing sheet production from calculated dimensions. The library generates technical documentation with precise dimensioning and annotation capabilities.

DXF Generation

Fabrication-friendly 2D top and side views created through ASCII generation without proprietary CAD APIs, ensuring compatibility with standard manufacturing equipment.

CSV Output Format

Bill of materials and cost outputs maintain simplicity, transparency, and ease of review through comma-separated values format.

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Material Properties Assumptions

Mild Steel Specifications

- Density: 7,850 kg/m³
- Yield strength: 250 MPa
- Platform thickness: 3.0 mm minimum

Stainless Steel Specifications

- Density: 8,000 kg/m³
- Yield strength: 215 MPa
- Platform thickness: 2.5 mm minimum

Safety Factor

Applied safety factor of 2.5 across all structural calculations to ensure adequate margin for manufacturing tolerances and operational variability.

Frame and Deck sizing assumptions

Payload Range	Tube Specification
Up to 300 kg	40×40×2.5 mm
Up to 700 kg	50×50×3.0 mm
Above 700 kg	60×60×4.0 mm

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Load Distribution Assumptions

Central Load Distribution

Platform load assumed to be centrally distributed across the entire surface area, representing worst-case scenario for frame stress analysis.

Wheel Load Equalization

Estimated wheel load calculated as equal distribution across wheel count, providing uniform support point analysis.

Static Loading Model

Dynamic and impact loads not explicitly modeled in current calculation methodology, focusing on static operational conditions.

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Cost and Timeline Estimation

Costing Methodology

- Material cost estimated using INR/kg assumptions
- Fabrication costs modeled by percentage plus fixed overhead
- Assembly costs calculated based on component complexity
- Planning-level estimates requiring supplier quotations for refinement

Timeline Calculation

- Manufacturing timeline estimated with load multipliers
- Size-based duration adjustments applied
- Fabrication and assembly phases separated
- Production routing data needed for precise scheduling

Timeline Alignment Note: Project assessment timeline of 72 hours represents development duration, while manufacturing timeline outputs represent fabrication and assembly estimates for the designed trolley.

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System Limitations

Understanding the current constraints ensures appropriate application and expectations for deliverable quality:

Structural Analysis Level

First-pass deterministic calculations, not finite element analysis. Results provide conceptual sizing guidance rather than certified structural validation.

CAD Output Scope

Concept-level models without detailed weld specifications, manufacturing tolerances, or standards-compliant drafting packages. Requires engineering review.

Drawing Generation Method

2D drawings generated from calculated dimensions rather than projected from full CAD assembly engine, limiting view consistency.

Cost and Timeline Precision

Planning-level estimates requiring refinement with supplier quotations and detailed production routing data before manufacturing commitment.

Workflow Automation

Semi-automated process still requiring manual review and final engineering sign-off before manufacturing release.

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